Formula Sheet MOOC | 45 Formulae | Weekly Divisions

Week 1: Sustainability Concepts

Environmental Efficiency (E-factor): Calculate the E-factor for a chemical production process to measure material efficiency and waste generation.

Formula: $\text{E-factor} = \frac{\text{Total Mass of Waste}}{\text{Total Mass of Product}}$ (Note: Total Mass of Waste = Total Mass of Inputs - Total Mass of Desired Product)

Units Required:

- **Mass of Waste:** kg (or grams)
- **Mass of Product:** kg (or grams)
- **E-factor:** dimensionless (kg waste / kg product)

Week 2: Measurements and Units

Equivalent Weight of H_2SO_4: What is the equivalent weight of H_2SO_4?

Answer: 49 g/eqv
Formula: $\mathrm{Eq.Wt.} = \frac{\mathrm{MW}}{\mathrm{n}}$

Units Required:

- **MW (Molecular Weight):** g/mol
- **n (n-factor/valency):** dimensionless (equivalents/mol)
- **Eq. Wt.:** g/eqv

Normality of H_2SO_4: What is the normality (N) of 1 M Solutions of H_2SO_4?

Answer: 2N

Formula: $\mathrm{N} = \mathrm{M} \times \mathrm{n}$

Units Required:

- **M (Molarity):** mol/L (M)
- **n (n-factor/valency):** equivalents/mol

- **N (Normality):** equivalents/L (N)

Molarity of $CaCO_3$: What is the molarity of 50 grams of calcium carbonate?

Answer: 0.5 moles/L

Formula: $\mathrm{Moles} = \frac{\mathrm{Mass}}{\mathrm{MW}}$

Units Required:

- **Mass:** grams (g)
- **MW (Molecular Weight):** g/mol

Molality of NaCl: Calculate the molality of 30.0 grams of NaCl dissolved in 500.0 mL of water.

Answer: 1.03 m

Formula: $\mathrm{m} = \frac{\mathrm{n_{solute}}}{\mathrm{Mass_{solvent}}}$

Units Required:

- $\mathrm{n_{solute}}$ **(Moles of solute):** mol
- $\mathrm{Mass_{solvent}}$**:** kg (You must convert mL/grams of water to kg)
- **m (Molality):** mol/kg (m)

Partial Pressure Calculation: What is the partial pressure of CO_2 if its concentration is 420 ppmv and atmospheric pressure is 1 atm?

Answer: 4.2×10^{4} atm

Formula: $\mathrm{P_i} = \left(\frac{\mathrm{C}}{10^6}\right) \times \mathrm{P_{total}}$

Units Required:

- **C (Concentration):** ppmv (parts per million by volume)
- $\mathrm{P_{total}}$**:** atm (or Pascals/Bar, but must match the output unit)
- $\mathrm{P_i}$ **(Partial Pressure):** atm

Total Dissolved Solids (TDS): Total Solids = 250 mg/L, Total Suspended Solids = 190 mg/L. What is the concentration of TDS?

Answer: 60 mg/L

Formula: $\mathrm{TDS} = \mathrm{TS} - \mathrm{TSS}$

Units Required:

- **TS (Total Solids):** mg/L

- **TSS (Total Suspended Solids):** mg/L
- **TDS:** mg/L

Upper Confidence Limit (UCL): Calculate the conservative estimate of the mean that accounts for data uncertainty (e.g., at 95% confidence).

Formula: $\mathrm{UCL} = \bar{x} + t\left(\frac{s}{\sqrt{n}}\right)$

Units Required:

- $\bar{x}$ **(Sample Mean):** Same units as the dataset (e.g., mg/L)
- t **(t-statistic):** dimensionless (derived from a lookup table based on confidence level and degrees of freedom)
- s **(Standard Deviation):** Same units as the dataset
- n **(Number of samples):** dimensionless count
- **UCL:** Same units as the dataset

Week 3: Ecology and Population

pH to Ion Concentration: Find the hydrogen ion concentration and the hydroxide ions concentration in tomato juice having pH of 4.1.

Answer: 7.94 × 10 ^{5} mol/L & 1.26 × 10 ^{10} mol/L.

Formula: $[H^+] = 10^{-pH}$ and $[OH^-] = \frac{10^{-14}}{[H^+]}$

Units Required:

- **pH:** dimensionless
- $[H^+]$ **and** $[OH^-]$**:** mol/L (Molarity)

Radioactive Decay (Half-Life): Calculate the concentration of a radioactive isotope (like Cesium) after a certain number of years given its half-life.

Formula: $k = \frac{0.693}{t_{1/2}}$ followed by $C_t = C_0 e^{-kt}$

Units Required:

- $t_{1/2}$ **(Half-life):** years (or days/seconds)
- k **(Decay Rate Constant):** years ^{1}
- C_0 **and** C_t **(Concentrations):** Becquerels (Bq) per kg, or any matching concentration unit.
- t **(Time):** years (must match the unit of k)

Week 4: Mass Balance and Reactors

CMBR Retention Time / Flow Rate: Calculate retention times in the CMBR operating at a flow rate of 50 m^3/day. Volume is 500 m^3.

Answer: 10 days

Formula: $\tau = \frac{V}{Q}$

Units Required:

- **V (Volume):** m^3 (or Liters, must match Q)
- **Q (Flow Rate):** m^3/day (or L/day)
- **τ (Retention Time):** days

CSTR Steady-State (First-Order):

Formula: $C_{out} = \dfrac{C_0}{1 + k\theta} = \dfrac{QC_0}{Q + kV}$

Units: C_0, C_{out}: mg/L | k: day^{-1} | $\theta = V/Q$: days | Q: m^3/day | V : m^3

CSTR Steady-State (Zero-Order):

Formula: $C_{out} = C_0 - k\theta = C_0 - \dfrac{kV}{Q}$

Units: Same as above; k: mg/L/day

Batch Reactor — Zero-Order:

Formula: $C_e = C_0 - kt$

Units: C_0, C_e: mg/L | k: mg/L/day | t: days (must match k)

Batch Reactor — Second-Order:

Formula: $\dfrac{1}{C_e} = \dfrac{1}{C_0} + kt$

Units: C_0, C_e: mg/L | k: L/(mg·day) | t: days

Stokes' Law — Force Balance on Settling Particle:

Force	Formula	Direction

Gravity	$F_g = \rho_p \frac{\pi}{6} D_p^3 g$	Downward
Buoyancy	$F_B = \rho_f \frac{\pi}{6} D_p^3 g$	Upward
Drag	$F_D = 3\pi\mu D_p v_r$	Upward

At terminal velocity: $F_g = F_B + F_D$ net force = 0

Units: ρ_p, ρ_f: kg/m^3 | D_p: m | μ: Pa·s | v_r: m/s | Forces: N

Batch Reactor Reaction Time: Time to reduce concentration of pollutant A to 1% of its initial value (k=0.23 min ^{1}).

Answer: 20 min

Formula: $\ln(\frac{C}{C_0}) = -kt$

Units Required:

- **C and C_0:** Must be in identical units (e.g., mg/L, %, fractions) so they cancel out.
- **k (Rate Constant):** min ^{1} or day ^{1}
- **t (Time):** Minutes or days (must match the time unit of k)

PFR Volume Calculation: Determine volume of Plug flow reactor for effluent of 50 mg/L (Q=100 m^3/day, k=0.23/day, Influent=200 mg/L).

Answer: 603 m^3

Formula: $\tau = \frac{-1}{k} \ln(\frac{C_{out}}{C_{in}})$ followed by $V = Q \times \tau$

Units Required:

- C_{out} **and** C_{in}**:** mg/L
- **k:** day ^{1}
- **τ:** days
- **Q:** m^3/day
- **V:** m^3

Steady State Accumulation
Concept: At steady state, the accumulation term in a control volume mass balance is exactly zero.

Formula: $\text{Accumulation} = \text{Input} - \text{Output} \pm \text{Reaction}$

Units: Typically expressed in terms of mass flux (e.g., kg/day or g/min), but in steady-state scenarios, the left side of the equation ($\frac{dM}{dt}$) simply becomes 0.

Simple Stream Mixing (Conservative Mass Balance): A wastewater stream mixes with a river. What is the downstream concentration of a conservative pollutant?

Formula: $C_{mix} = \frac{Q_1 C_1 + Q_2 C_2}{Q_1 + Q_2}$

Units Required:

- Q_1, Q_2 **(Flow Rates):** Must be identical (e.g., m³/s or MLD)
- C_1, C_2 **(Concentrations):** Must be identical (e.g., mg/L)
- C_{mix}: mg/L (Matches input concentration units)

Week 5: Oxygen Demand and Nutrient Cycles

BOD Calculation: A 2% solution of a sewage sample is incubated for 5 days. Depletion of oxygen was 4 ppm. BOD?

Answer: 200 mg/L

Formula: $BOD = \frac{DO_{initial} - DO_{final}}{P}$

Units Required:

- **DO (Dissolved Oxygen):** mg/L (Note: ppm in water is equivalent to mg/L)
- **P (Dilution Factor):** dimensionless fraction (e.g., 2% = 0.02)
- **BOD:** mg/L

Theoretical Oxygen Demand (ThOD): Compute the theoretical oxygen demand of water containing glucose ($C_6H_{12}O_6$) expressed as mg of O_2 / mg of glucose.

Answer: 1.07

Formula: $ThOD = \frac{\text{MW of } O_2 \text{ required}}{\text{MW of Compound}}$ (Requires writing the balanced oxidation reaction first, e.g., $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$)

Units Required:

- **MW (Molecular Weight):** g/mol
- **ThOD:** mg/L or ratio (mg O_2 / mg substance)

BOD Kinetics (Ultimate BOD): Given the ultimate BOD (L_0) and rate constant (k), calculate the BOD exerted after t days. *(Note: Your assignments only tested the lab dilution formula, but this kinetic formula is a guaranteed exam target).*

Formula: $\mathrm{BOD_t} = \mathrm{L_0}(1 - \mathrm{e}^{-\mathrm{kt}})$

Units Required:

- $\mathrm{L_0}$ **(Ultimate BOD):** mg/L
- k **(Deoxygenation Rate Constant):** day ¹ (Base e. If given in Base 10 as K, the formula becomes $1 - 10^{-\mathrm{Kt}}$)
- t **(Time):** days
- $\mathrm{BOD_t}$**:** mg/L

Minimum Detectable BOD:

Formula: $\mathrm{BOD_{min}} = \dfrac{\Delta\mathrm{DO_{min}}}{\mathrm{p}}$

Units: $\Delta\mathrm{DO_{min}}$: mg/L (typically ≥ 1.5 mg/L) | p: dimensionless dilution fraction | $\mathrm{BOD_{min}}$: mg/L

> *Example (Assignment Q9): p = 10/300, ΔDO_min = 1.5   BOD_min = 1.5 × 30 = 45 mg/L*

Ultimate BOD from BOD₅:

Formula: $\mathrm{L_0} = \dfrac{\mathrm{BOD_5}}{1 - \mathrm{e}^{-5\mathrm{k}}}$

Units: $\mathrm{BOD_5}$: mg/L | k: day ¹ (base e) | $\mathrm{L_0}$: mg/L

Week 6: Environmental Risk and EIA

Hazard Quotient (HQ) for Non-Cancer Risk: Used to determine if exposure to a chemical contaminant exceeds safe levels.

Formula: $\mathrm{HQ} = \frac{\mathrm{CDI}}{\mathrm{RfD}}$ (Where CDI is Chronic Daily Intake and RfD is Reference Dose)

Units Required:

- CDI**:** mg/kg-day (milligrams of pollutant per kg of body weight per day)
- RfD**:** mg/kg-day
- HQ**:** dimensionless (If HQ > 1, there is a potential health risk)

Cancer Risk (CR):

Formula: $\mathrm{CR} = \mathrm{CDI} \times \mathrm{SF}$

Units: CDI: mg/kg-day | SF: (mg/kg-day) ¹ | CR: dimensionless probability

> *CR > 10 ^{4} unacceptable. CR < 10 ^{6} negligible.*

Week 7: Water Quantity and Quality

Storm Water Runoff (Rational Method): 150 ha catchment: 40% paved (C=0.85), 35% residential (C=0.45), 25% lawns (C=0.20). Rainfall 50 mm/h.

Answer: 10.4 m^3/s

Formula: $Q = \frac{C \cdot i \cdot A}{360}$

Units Required:

- **C (Runoff Coefficient):** dimensionless
- **i (Rainfall Intensity):** mm/h (Must be mm/h for the 360 denominator to work)
- **A (Area):** hectares (ha)
- **Q (Discharge):** m^3/s

Fire Demand: Population 10,00,000 using NBFU formula.

Answer: 145 MLD

Formula: $Q = 4637\sqrt{P}(1 - 0.01\sqrt{P})$

Units Required:

- **P:** Population in thousands (e.g., for 1,000,000 people, P = 1000)
- **Q:** L/min (You must multiply by 60 and 24 to get L/day, then divide by 10^6 for MLD)

Pipe Diameter Calculation (Hazen-Williams variant): 1 lakh population, 200 lpcd, 80 km length, 10 m elevation diff, f=0.003.

Answer: 0.64 m

Formula: $h_f = \frac{fLQ^2}{12.1D^5}$

Units Required:

- h_f **(Head Loss):** meters (m)
- **f (Friction Factor):** dimensionless
- **L (Length):** meters (m) (Convert 80 km to 80,000 m)
- **Q (Flow Rate):** m^3/s (Convert population × lpcd into m^3/s)
- **D (Diameter):** meters (m)

Total Hardness: Cations: Ca^{2+}=60, Mg^{2+}=36. Compute total hardness as $CaCO_3$.

Answer: 300 mg/L

Formula: $\mathrm{TH} = [\mathrm{Ca}^{2+}] \times (\frac{50}{20}) + [\mathrm{Mg}^{2+}] \times (\frac{50}{12.2})$

Units Required:

- **Ion Concentrations:** mg/L
- **Multiplier:** Ratio of equivalent weights (50 is Eq. Wt. of $CaCO_3$; 20 is Eq. Wt. of Ca; 12.2 is Eq. Wt. of Mg)
- **TH:** mg/L as $CaCO_3$

Most Probable Number (MPN) - Thomas Equation: Calculate the bacterial density of coliforms per 100 mL of water from multiple tube fermentation test results.

Formula: $\mathrm{MPN/100\ mL} = \frac{\text{Number of Positive Tubes} \times 100}{\sqrt{(\text{mL sample in negative tubes}) \times (\text{mL sample in all tubes})}}$

Units Required:

- **Number of Positive Tubes:** dimensionless count
- **Volumes:** mL (Must strictly use the volume of the original sample present in the tubes)
- **MPN:** organisms / 100 mL

Carbonate & Non-Carbonate Hardness: $\mathrm{HCO_3^-}$ = 244 mg/L. Total alkalinity is 200 mg/L. TH is 300 mg/L.

Answer: CH = 200 mg/L. NCH = 100 mg/L.

Formula: $\mathrm{CH} = \min(\mathrm{TH}, \mathrm{Alkalinity})$ and $\mathrm{NCH} = \mathrm{TH} - \mathrm{CH}$

Units Required:

- **TH, Alkalinity, CH, NCH:** mg/L as $\mathrm{CaCO_3}$

Arithmetic Population Forecasting: The population of a town increased by 18,000; 22,000; and 20,000 in the last three decades. Present population is 3,00,000. Population after 2 decades will be:

Answer: 3.40 lakh

Formula: $\mathrm{P_n} = \mathrm{P_0} + \mathrm{n\bar{x}}$

Units Required:

- $\mathrm{P_0}$ **(Present Pop):** Persons
- **n:** Number of decades
- $\bar{x}$**:** Average arithmetic increase per decade

Geometric Population Forecasting: The arithmetic method was in the 2019 assignment, which means the geometric method is highly likely to appear on the final.

Formula: $\mathrm{P_n = P_0 \left(1 + \frac{r}{100}\right)^n}$

Units Required:

- $\mathrm{P_0}$ **(Present Population):** Persons
- r **(Assumed uniform percentage growth rate):** % per decade (or per year)
- n **(Number of terms/decades):** Must match the time unit of r
- $\mathrm{P_n}$**:** Persons

Incremental Increase Population Forecasting: The third major population forecasting method (alongside Arithmetic and Geometric) that accounts for a continuously changing growth rate.

Formula: $\mathrm{P_n = P_0 + n\bar{x} + \frac{n(n+1)}{2}\bar{y}}$

Units Required:

- $\mathrm{P_0}$**:** Persons (Present Population)
- n**:** Number of decades
- $\mathrm{\bar{x}}$**:** Persons (Average arithmetic increase per decade)
- $\mathrm{\bar{y}}$**:** Persons (Average incremental increase—the change in the increase—per decade)
- $\mathrm{P_n}$**:** Persons

Week 8: Water Treatment

Filter Area: Handle 2 MLD of water with rate of 200 lit/hr/m².

Answer: 417 m²

Formula: $\mathrm{Area = \frac{Q}{Rate}}$

Units Required:

- **Q:** L/hr or L/day (Must be converted from MLD to match the time unit of the Rate)
- **Rate:** L/hr/m² or L/day/m²
- **Area:** m²

Settling Velocity (Stokes Law): diameter=0.06mm, G=2.65, T=25°C, kinematic viscosity=0.897 cSt.

Answer: 3.6 mm/s

Formula: $v_s = \frac{g(G-1)d^2}{18\nu}$

Units Required:

- **g (Gravity):** 9.81 m/s²
- **G (Specific Gravity):** dimensionless
- **d (Diameter):** meters (Convert 0.06 mm to 0.06×10^{-3} m)
- **ν (Kinematic Viscosity):** m²/s (Convert cSt: $1\text{cSt} = 10^{-6}$ m²/s)
- **v_s (Velocity):** m/s (Convert final answer back to mm/s if options require it)

Chlorine Demand & Bleaching Powder: Dose=0.8, residual=0.3. Bleaching powder is 20% Cl_2.

Answer: Demand = 0.5 mg/L, Powder = 4 mg/L

Formula: $\text{Demand} = \text{Dose} - \text{Residual}$ and $P\text{owder} = \frac{\text{Dose}}{\%\text{Available}}$

Units Required:

- **Dose, Residual, Demand:** mg/L
- **% Available:** decimal fraction (e.g., 20% = 0.20)

Sedimentation Efficiency: V_s = 17.45 m/day, Overflow = 30 m³/day/m².

Answer: 58.27%

Formula: $\eta = \left(\frac{V_s}{V_o}\right) \times 100$

Units Required:

- V_s **(Settling Velocity):** m/day (Must match V_o)
- V_o **(Overflow Rate):** m³/day/m² (which simplifies to m/day)
- η **(Efficiency):** %

Backwash Volume: A Rapid Sand Filter treats 30 MLD... Backwash rate is 6 times the filtration rate and lasts for 10 minutes.

Answer: 249,960 L

Formula: $\text{Volume} = \text{Area per unit} \times \text{Backwash Rate} \times \text{Time}$

Units Required:

- **Area:** m²
- **Backwash Rate:** L/h/m²
- **Time:** hours (Convert from minutes to match the rate)
- **Volume:** Liters (L)

Surface Overflow Rate (SOR): Your assignments tested Sedimentation Efficiency using V_o, but you may be asked to calculate V_o from scratch.

Formula: $V_o = \frac{Q}{A_{surface}}$ (where $A_{surface} = \text{Length} \times \text{Width}$ of the tank)

Units Required:

- Q **(Flow Rate):** m^3/day
- $A_{surface}$ **(Surface Area):** m^2
- V_o **(Overflow Rate):** m^3/day/m^2 (which mathematically simplifies to m/day, allowing you to compare it directly to Settling Velocity V_s)

Disinfection Kinetics (Chick's Law): Calculates the time required to kill a specific percentage of pathogens using a disinfectant like chlorine.

Formula: $\ln\left(\frac{N_t}{N_0}\right) = -kt$

Units Required:

- N_t**:** Pathogen concentration at time t (e.g., organisms/100mL)
- N_0**:** Initial pathogen concentration (e.g., organisms/100mL)
- k**:** Disinfection rate constant (time^{-1}, e.g., min^{-1})
- t**:** Time (Must match the time unit of k, e.g., minutes)

Week 9: Wastewater Treatment Concepts

Trickling Filter Efficiency (NRC Formula): Calculate the BOD removal efficiency of a single-stage trickling filter.

Formula: $\eta = \frac{100}{1+0.0044\sqrt{\frac{W}{V}}}$ *(Note: $W = Q \times y_i$, which is the total BOD loading)*

Units Required:

- η **(Efficiency):** %
- W **(BOD Loading):** kg/day
- Q **(Flow Rate):** ML/day (or m^3/day, ensure it matches loading calculation)
- y_i **(Influent BOD):** mg/L
- V **(Volume of Filter):** hectare-meters (ha-m) or m^3 depending on the specific constant used (The 0.0044 constant requires W in kg/day and V in ha-m).

Activated Sludge Process - F/M Ratio: The Food-to-Microorganism ratio, a critical design parameter to ensure bacteria have the right amount of organic matter to consume in the aeration tank.

Formula: $\frac{F}{M} = \frac{Q \times S_0}{V \times MLSS}$

Units Required:

- Q **(Flow Rate):** m³/day or MLD
- S_0 **(Influent BOD):** mg/L
- V **(Aeration Tank Volume):** m³ or ML (Must match Q volume unit)
- $MLSS$ **(Mixed Liquor Suspended Solids):** mg/L
- **F/M Ratio:** day ^{1} (or kg BOD / kg MLSS-day)

Week 10: Solid Waste Management

Sludge Volume Index (SVI): MLSS = 3000 mg/L, settled volume (SV_{30}) = 300 mL/L.

Answer: 100 mL/g

Formula: $SVI = \frac{SV_{30} \times 1000}{MLSS}$

Units Required:

- SV_{30}**:** mL/L (Volume of settled sludge per Liter of mixed liquor)
- **MLSS:** mg/L
- **SVI:** mL/g

Daily BOD Load: Population 35,000, 180 lpcd, BOD is 220 mg/L.

Answer: 1,386 kg/day

Formula: $\text{Load} = Q \times BOD \times 10^{-6}$

Units Required:

- **Q:** L/day (Population × lpcd)
- **BOD:** mg/L
- **Load:** kg/day (The 10^{-6} converts mg to kg)

Sludge Volume Reduction: Moisture drops from 98% to 92%.

Answer: 75%

Formula: $\frac{V_1}{V_2} = \frac{S_2}{S_1}$

Units Required:

- **S (Percent Solids):** % (Calculated as $100 - \%Moisture$)
- **V (Volume):** Any unit, as long as it forms a ratio.

Mixed Waste Weighted Averages (Energy/Moisture): 50% organic (4000 kcal/kg), 50% inert (0 kcal/kg).

Answer: 2000 kcal/kg

Formula: $E_{mixed} = \sum(Fraction_i \times Value_i)$

Units Required:

- **Fraction:** Decimal percentage (e.g., 50% = 0.50)
- **Value:** kcal/kg (for energy) or % (for moisture content)

Density of Mixed Solid Waste: Calculates the overall density of a garbage truck or landfill pile based on the mass fractions and individual densities of paper, glass, food, etc.

Formula: $\rho_{mix} = \frac{100}{\sum\left(\frac{P_i}{\rho_i}\right)}$

Units Required:

- P_i **(Percentage):** % by mass of each component (e.g., 20 for 20%)
- ρ_i **(Density):** kg/m³ (Density of that specific component)
- ρ_{mix}: kg/m³

Week 11: Air and Noise Pollution

Gaussian Dispersion Model (Centerline): 2.4 g/min SO_2, 15m stack, 3 m/s wind, σy=25m, σz=15m.

Answer: 6.9 μg/m³

Formula: $C = \frac{Q}{\pi u \sigma_y \sigma_z}$ (For ground level centerline $y = 0, z = 0$)

Units Required:

- **Q (Emission Rate):** g/s (Convert 2.4 g/min to 0.04 g/s)
- **u (Wind Speed):** m/s
- σ_y, σ_z **(Dispersion Coefficients):** meters (m)
- **C:** g/m³ (Multiply by 10^6 to convert to μg/m³)

Concentration Conversion: Convert 10 ppm CO to μg/m³ at 25°C and 1 atm. (MW=28).

Answer: 11,450 μg/m³

Formula: $C = \frac{ppm \times MW \times 1000}{V_m}$

Units Required:

- **ppm:** parts per million
- **MW:** g/mol
- V_m **(Molar Volume):** L/mol (At 25°C, $V_m = 24.45$ L/mol)
- **C:** μg/m³

Decibel Addition: Two identical noise sources of 50 dB each.

Answer: 53 dB

Formula: $L_T = 10\log_{10}(\sum 10^{L_i/10})$

Units Required:

- L_i**:** Individual Decibel levels (dB)
- L_T**:** Total Decibel level (dB)

Plume Rise: Effective stack height $H = h_s + \Delta h$

Formula: $\Delta h = \frac{vd}{u}\left[1.5 + 2.68 \times 10^{-3} P \left(\frac{T_s - T_a}{T_a}\right) d\right]$

Units: v: m/s (exit velocity) | d: m (inner diameter) | u: m/s (wind speed) | P: mb (pressure) | T_s, T_a: K | Δh: m

Full 3D Gaussian Dispersion:

$$C(x,y,z) = \frac{S}{2\pi u \sigma_y \sigma_z} \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \exp\left(-\frac{(z-H)^2}{2\sigma_z^2}\right)$$

Ground-level ($z = 0$):

$$C(x,y,0) = \frac{S}{\pi u \sigma_y \sigma_z} \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \exp\left(-\frac{H^2}{2\sigma_z^2}\right)$$

Units: S: g/s | u: m/s | σ_y, σ_z, H: m | C: g/m³ (×10⁶ for μg/m³)

> *The ground-level coefficient is* π *(not* 2π*) because perfect ground reflection doubles concentration at* $z = 0$ *— the ground acts as a mirror image source, so* $2 \times \frac{S}{2\pi\cdots} = \frac{S}{\pi\cdots}$.

Week 12: Key Constants (case-study week — no numeric formulas)

Fact	Value
Total global freshwater	2.5% of all water
Freshwater in glaciers/ice caps	68.7% of freshwater

Freshwater as groundwater	30.1% of freshwater
Surface water (lakes, rivers, swamps)	0.3% of freshwater
Rivers alone	0.006% of freshwater
Cape Town Level 6B daily limit	50 L/person/day
Floating marine debris that is plastic	90%
Marine litter from land-based sources	80%
Bhopal MIC leak date	December 3, 1984
Bhopal confirmed deaths	3,787
Bhopal total injuries	558,125